

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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Flow

- Announcement
- Recap
- Module-2 starts

Announcement

- Lectures
 - No class on next Tuesday (29th Sep)
 - Class tomorrow (Friday 25th Sep) at 10 am
- Check GitHub regularly

3

Course Flow

Module 1: **Probability** – Understanding Uncertainty



Module 2: **Statistics** – Learning from Data



Module 3: **Reliability** – Decision under Uncertainty

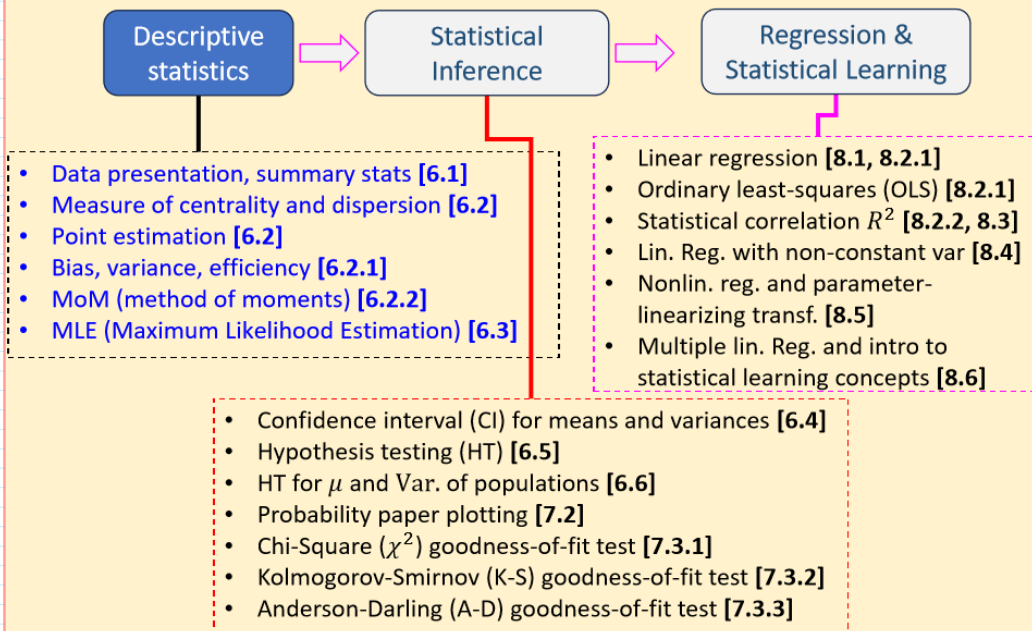


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5

Module II: Chapter 1 (M2C1)

Module 2: **Statistics** – Learning from Data



M2C1

Point Estimation Framework: $\hat{\theta}$ to characterize unknown population parameters from finite empirical observations.

Properties of Estimators: unbiasedness, efficiency, consistency, and variance bounds to evaluate optimal estimator trade-offs.

Method of Moments (MoM): parameter estimates by equating theoretical population moments to corresponding sample moments.

Maximum Likelihood Estimation (MLE): construct likelihood and log-likelihood functions to estimate parameter values that maximize the probability of observed sample data.

Sampling Distributions & Standard Errors: Quantify the variability of point statistics.

Motivation

How to extract single best-estimate model parameters from noisy, finite field measurements?

Bias and precision

N.T: How close the dart throws are to the bullseye?

• Accuracy/bias:

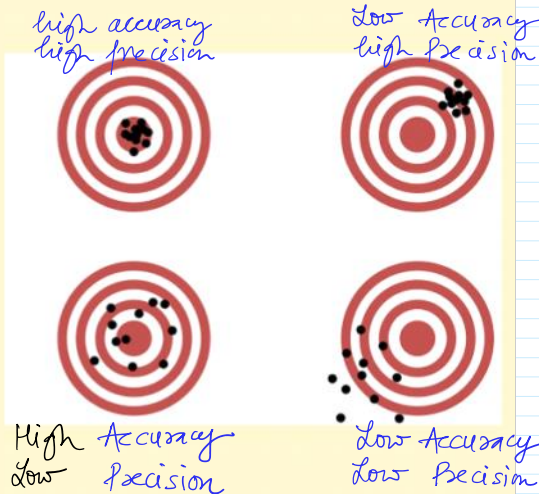
→ closeness to center, on an average

→ Neutrality/Unbiasedness

• Precision:

How close are the dart throws to each other?

→ Consistency/Repeatability



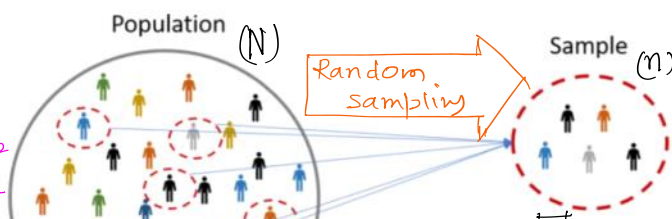
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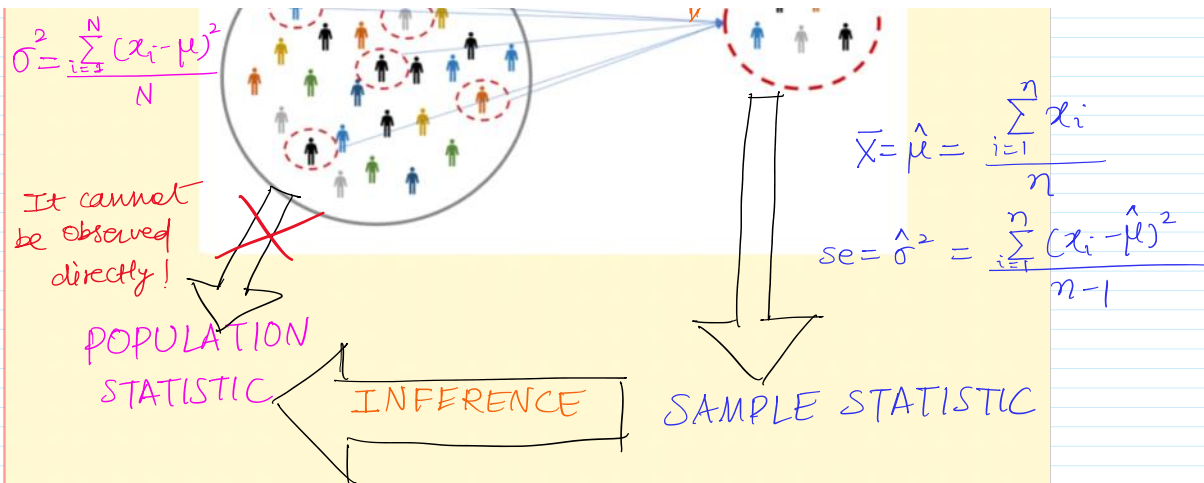
Random Sampling

e.g., Election polls, Mechanical properties

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$





9

Parameter estimation: method of moments

Given: 1) Data & Probability Model, to estimate parameters of the

- Obtain the moments (e.g., mean, variance) from two sources:

- Observations:

$$\hat{\mu} = E[X] =$$

$$\hat{\sigma}^2 = E[(X - \mu)^2] =$$

{ mean } of observed data

{ stdev }

- Parameters:

reqn. for { mean } of a PDF

{ stdev }

For point estimate,

- Equate them to solve for the parameters

10

Distributions and parameters

Dist.	PDF	CDF	Mean & Variance
Normal/ Gaussian	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$	$F(x) = \Phi(x)$ Use z-table	$E[X] = \mu$ $\text{Var}[X] = \sigma^2$
Uniform	$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$	$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b] \\ 1 & \text{otherwise} \end{cases}$	$E[X] = \frac{1}{2}(a+b)$ $\text{Var}[X] = \frac{1}{12}(b-a)^2$
Exponential	$f(x) = \lambda \exp(-\lambda x)$	$F(x) = 1 - \exp(-\lambda x)$	$E[X] = \frac{1}{\lambda}$ $\text{Var}[X] = \frac{1}{\lambda^2}$
Gumbel	$f(x) = \frac{1}{\beta} \exp[-z + \exp(-z)]$, where $z = \frac{x-\alpha}{\beta}$	$F(x) = \exp[-\exp(-z)]$	$E[X] = \alpha + \beta\gamma$ $\text{Var}[X] = \frac{\pi^2}{6}\beta^2$ $\gamma \approx 0.577$
Lognormal	$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2\right]$	$F(x) = \Phi\left(\frac{\ln x - \mu}{\sigma}\right)$ Use z-table	$E[X] = \exp\left(\mu + \frac{\sigma^2}{2}\right)$ $\text{Var}[X] = [\exp(\sigma^2) - 1] \exp(2\mu + \sigma^2)$

Example

- Assume that Earthquakes in Rome expressed as MCS index follow Gumbel type-I distribution. Earthquake intensity and number of earthquakes between 1000 CE – 1980 CE are:

MCS intensity, x_i	2	3	4	5	6	7	$\sum x = \sum x_i f_i$	$\sum x^2 = \sum x_i^2 f_i$	$\sum f_i$
Number, f_i	113	132	56	22	4	2	994	3328	329

Find the parameters of the distribution.

- Observations:

$$E[X] = \mu_X = 994/329 \approx 3.02$$

$$\text{Var}[X] = E[X^2] - (E[X])^2 \approx 0.99$$

- Parameters for Gumbel type-I:

$$E[\cdot] = \alpha + \beta\gamma$$

$$\text{Var}[\cdot] = \frac{\pi^2}{6}\beta^2$$

$$\left\{ \text{MoM} \right\} \Rightarrow \beta \approx 0.77$$

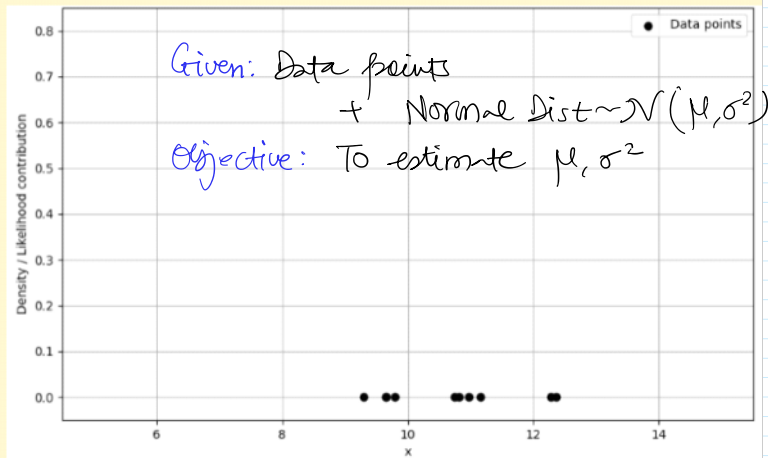
$$\alpha \approx$$

- Equate them to solve for the parameters

Maximum likelihood estimation

Basic idea

Objective of MLE:



Maximum likelihood estimation

Define likelihood,

$$\mathcal{L}(\mu, \sigma | x_i) = f_x(x_i | \mu, \sigma)$$

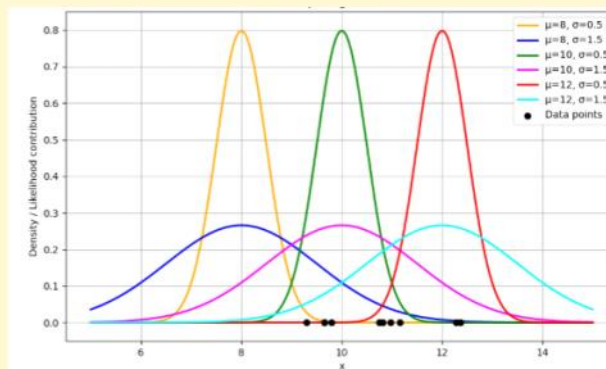
$$\mathcal{L}(\mu, \sigma | \underline{x}) = \prod_{i=1}^n f_x(x_i | \mu, \sigma)$$

$$\hat{\mu}, \hat{\sigma} =$$

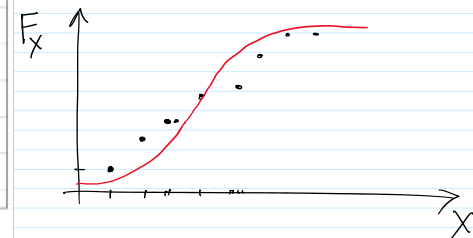
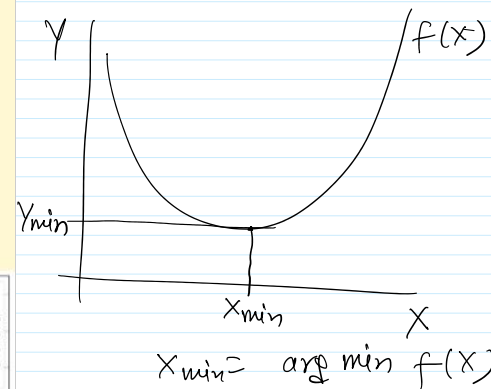
$$\operatorname{argmax} \mathcal{L}(\mu, \sigma | \underline{x})$$

$$\hat{\mu}, \hat{\sigma} =$$

$$\operatorname{argmax} \ln \mathcal{L}(\mu, \sigma | \underline{x})$$



$$\underline{x} = \{x_1, x_2, \dots, x_n\}$$



Maximum likelihood estimation

Find $\mathcal{L}(\mu, \sigma | x_i) = f(x_i | \mu, \sigma)$ for each dart, x_i .

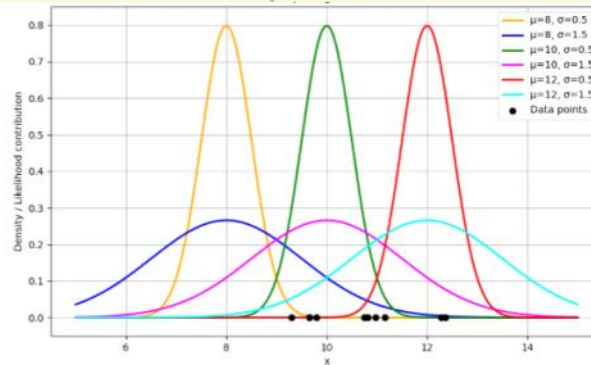
Maximize the product of likelihood,

$$\mathcal{L}(\mu, \sigma | \mathbf{x}) = \prod_i^n f(x_i | \mu, \sigma)$$

$$\hat{\mu}, \hat{\sigma} = \arg \max_{\mu, \sigma} \mathcal{L}(\mu, \sigma | \mathbf{x})$$

$$\hat{\mu}, \hat{\sigma} = \arg \max_{\mu, \sigma} \ln[\mathcal{L}(\mu, \sigma | \mathbf{x})]$$

Excel Demo!



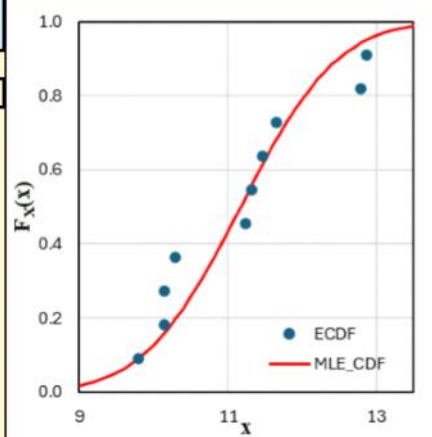
color	oran.	blue	green	mag.	red	turq.
μ_{guess}	8	8	10	10	12	12
σ_{guess}	0.5	1.5	0.5	1.5	0.5	1.5
x_i	Likelihood					
10.7	0.00	0.05	0.26	0.24	0.03	0.19
9.8	0.00	0.13	0.73	0.26	0.00	0.09
11.0	0.00	0.04	0.12	0.22	0.10	0.21
12.3	0.00	0.00	0.00	0.08	0.68	0.26
9.6	0.00	0.15	0.62	0.26	0.00	0.08
9.6	0.00	0.15	0.62	0.26	0.00	0.08
12.4	0.00	0.00	0.00	0.08	0.61	0.26
11.2	0.00	0.03	0.06	0.20	0.19	0.23
9.3	0.03	0.18	0.30	0.24	0.00	0.05
10.8	0.00	0.05	0.21	0.23	0.05	0.19
Sum-log-lik.	-166	-31	-32	-17	-59	-20

$\mu_{\text{goal-seek}}$ 10.67
 $\sigma_{\text{goal-seek}}$ 1.03

Lik.	Log-lik.
0.39	-0.4
0.27	-0.6
0.37	-0.4
0.11	-0.9
0.24	-0.6
0.24	-0.6
0.10	-1.0
0.35	-0.5
0.16	-0.8
0.38	-0.4

Use "solver" to maximize Sum-log-lik. by changing μ, σ .

Sum-log-lik. -6.3



M2C2

Interval Estimation: two-sided and one-sided CI for population μ and σ^2 under known and unknown variance assumptions.

Hypothesis Testing Principles: H_0 and H_1 .

Error Analysis & Test Power: Type I (α) and Type II (β) error probabilities, p -values.

Parametric Testing: Z -tests, t -tests, χ^2 -tests, and F -tests for single-sample and two-sample population parameters.

Empirical Probability Plotting: probability paper and ECDFs to evaluate candidate dist. visually.

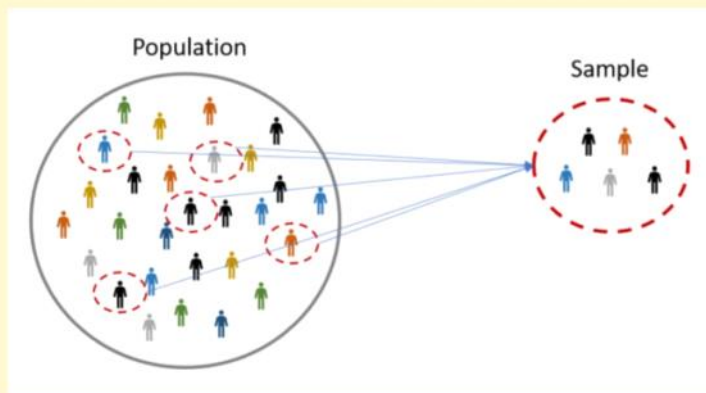
Goodness-of-Fit Verification: χ^2 , Kolmogorov-Smirnov (K-S), and Anderson-Darling (A-D) non-parametric hypothesis tests to validate distribution fits.

Motivation

How to test hypotheses and validate probability models?

Point Estimates $\Rightarrow$ Confidence Interval

- Issue with the point estimate, $\hat{\mu}$
 - $\rightarrow$ No information on how close $\hat{\mu}$ is to μ .
 - $\rightarrow$ Different sampling, different estimate



18

Distribution of Mean

- Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables having a distribution with mean μ and var. σ^2 . If Y is the sample average of X_i 's.

Find μ_Y and σ_Y^2

$$Y = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} X_1 + \frac{1}{n} X_2 + \dots + \frac{1}{n} X_n$$

$$\mu_Y = \frac{1}{n} \mu_1 + \frac{1}{n} \mu_2 + \dots + \frac{1}{n} \mu_n$$

Since $\mu_1 = \mu_2 = \dots = \mu_n = \mu$

$$\Rightarrow \boxed{\mu_Y = \mu} \quad \boxed{\sigma_Y^2 = \sigma^2/n}$$

CLT

19

Confidence interval

- Basic idea:

POPULATION MEAN μ is $\bar{X}$ (SAMPLE MEAN).
Best estimate of μ is $\bar{X}$.

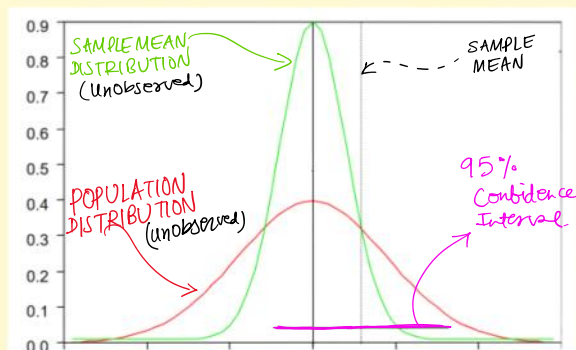
→ But $\bar{X}$ will never be equal to μ .

→ So, we can form an interval around $\bar{X}$ that covers μ .

20

Confidence interval for μ

- Basic idea:



21

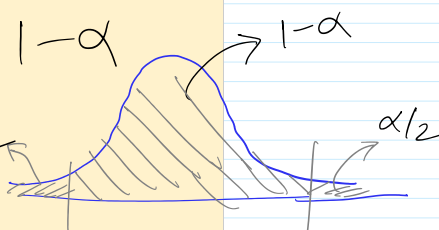
Confidence interval for μ

- $\bar{x}$: sample mean
- n : sample size
- α : significance level 5%
- $100(1 - \alpha)\%$: confidence level 95%

$100(1 - \alpha)\%$ confidence interval is

$$\Pr\left[-Z_{\alpha/2} \leq \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}\right] = 1 - \alpha$$

$$\Rightarrow \Pr\left[\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right] = 1 - \alpha$$



22

Example-1 on CI

- $n = 10$ measurements of impact energy (J) are as follows:
 - 64.1, 64.7, 64.5, 64.6, 64.5, 64.3, 64.6, 64.8, 64.2, 64.3
 - The sample mean is $\bar{x} = 64.46$
- Assume impact energy is normally distributed with $\sigma = 1$ J
- Find a 95% CI for μ , the mean impact energy

$$Z_{\alpha/2} \equiv Z_{0.025}$$

$$64.46 - 1.96 \times \frac{1}{\sqrt{10}} \leq \mu \leq 64.46 + 1.96 \times \frac{1}{\sqrt{10}}$$

$$63.84 \leq \mu \leq 65.08$$

23

Standard Normal Probabilities

Z-table

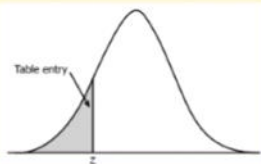


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Interpretation of a CI

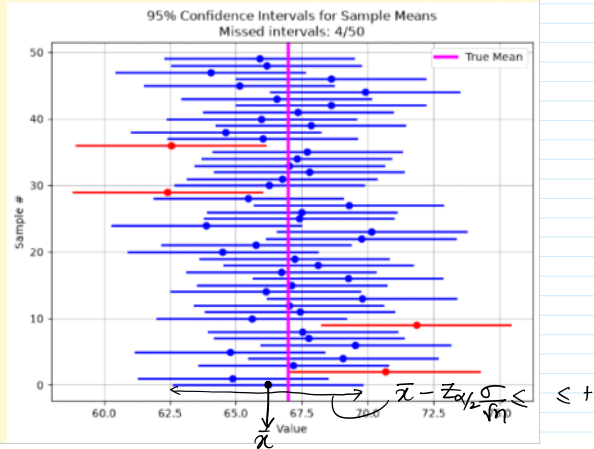
- ~~The true mean μ is within the CI with a probability of 95%.~~

Tempting BUT False ↗

- Each C.I. either contains μ or it does NOT

$$\Pr(\mu \in CI) = \begin{cases} 0 & \text{or} \\ 1 \end{cases}$$

- So then what was 95%?
- "CONFIDENCE level"



Choice of sample size

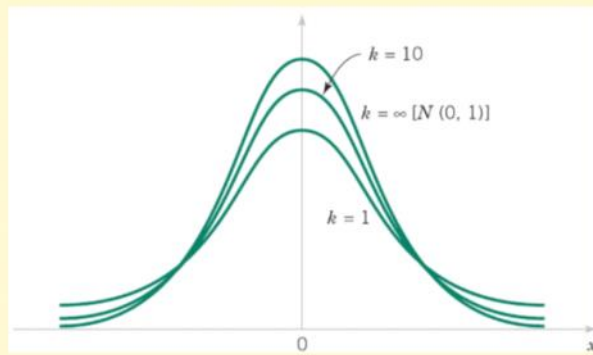
- For a target error in the length of interval

we want to be $100(1-\alpha)\%$ confident that the error $|\bar{X} - \mu|$ will not exceed E , what is the sample size n ?

$$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq E$$

$$\Rightarrow n \geq \left(z_{\alpha/2} \frac{\sigma}{E} \right)^2$$

Small sample sizes



27

Confidence interval for σ^2

- Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean μ and stdev σ . Let s^2 be the sample variance.

1

Example-3 on CI for σ^2

- $n = 10$ measurements of concrete cylinder compressive strength (in MPa) are recorded as follows:
32.1,33.4,31.8,32.5,33.0,32.7,31.9,33.2,32.4,32.0
- The sample variance is calculated as $s^2 = 0.33 \text{ MPa}^2$
- Assuming compressive strength is normally distributed, find a **95% confidence interval** for:
 - The population variance σ^2
 - The population standard deviation σ

2

Thank you!